

Test-time Adaptation for Foundation Medical Segmentation Model without Parametric Updates

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Abstract

- Propose TTA method for a foundation medical segmentation model (e.g., MedSAM) **without parametric updates**
- **Only the image embedding Z** is refined at test time
- Achieve about 3% Dice score improvements while reducing computational complexity by **over 7 times**

Update Type	Method	Datasets			GFLOPs↓
		OC	OD	SCGM	
	Direct Inference	87.77	80.57	56.67	-
Normalization	TENT [25] (ICLR'21)	86.24	78.08	54.34	743.5
	InTEnt [6] (CVPR'24)	86.95	82.40	49.81	744.8
	GraTa [32] (AAAI'25)	88.57	80.41	55.14	2975.6
	PASS [30] (TMI'24)	84.05	71.81	54.80	745.8
Full	MEMO [33] (NeurIPS'22)	88.12	80.04	57.48	5646.5
	CRF-SOD [24] (CVPR'23)	88.95	79.79	56.81	1115.2
Latent	<i>Ours</i>	90.84	84.19	59.49	97.3

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Why TTA is needed in Medical Segmentation

- Foundation segmentation models for medical images have attracted massive attention
 - Because medical image segmentation is essential for downstream clinical and analysis tasks
- MedSAM is the most popular foundation segmentation model and as a strong zero-shot segmenter
 - Strength: usually produce a high-quality RoI mask without task-specific training
 - Weakness
 - **Extreme modality imbalance** and less-represented modalities
 - Intricate structures and appearance: hard lesions/targets
 - **Bounding box prompt-induced perturbations**

Why TTA is needed in Medical Segmentation

- Why not simply fine-tune?
 - overwhelming **computational consumption** of fine-tuning
 - **inherent data scarcity and privacy concerns** for annotated medical images
- Moreover, even a well-calibrated and fine-tuned model can still **face test-time perturbations** (e.g., prompt shifts)

Why TTA is needed in Medical Segmentation

- TTA is feasible, but existing TTA **does not scale** well to foundation models
 - BN-centric objectives vs LN-based foundation models: limited update signals
 - Whole-model updates and **catastrophic forgetting**
 - **Computational complexity is ignored**, but it is critical here
- Propose **updates latent representations** rather than any model parameters

Why TTA is needed in Medical Segmentation

- Contribution
 1. Theory: show that direct latent refinement can approach the same goal as parametric updates under the MedSAM architecture, enabling high computational efficiency and segmentation performance without catastrophic forgetting
 2. Method: propose a distribution-approximated latent CRF loss based on a Maximum Mean Discrepancy (MMD)-aware co-occurrence measure to introduce useful cues from the input space and obtain sufficient statistics of image embedding over the input image.
 3. Experiments: validate on three multi-center datasets with strong appearance variations and simulated prompt shifts, reporting around 3% Dice improvements and about $7\times$ reduction in computational complexity versus existing methods

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Medical Image Segmentation with TTA

- The setting: distribution shifts during inference and the multi-center issue
- Optimization
 - BN-layer optimization (BN-centric fine-tuning)
 1. Updating the affine-transformation parameters of BN
 2. Leveraging source-trained BN statistics (often via sophisticated alignment)
 - fine-tuning the overall model (whole-model parametric updates)

Medical Image Segmentation with TTA

- Why parametric-update TTA does not scale to foundation segmentation models
 - Limited update signals under LN backbones
 - Catastrophic forgetting for large pre-trained models
 - Computational complexity is ignored during adaptation
- + Limited works specific to the TTA of foundation medical segmentation models

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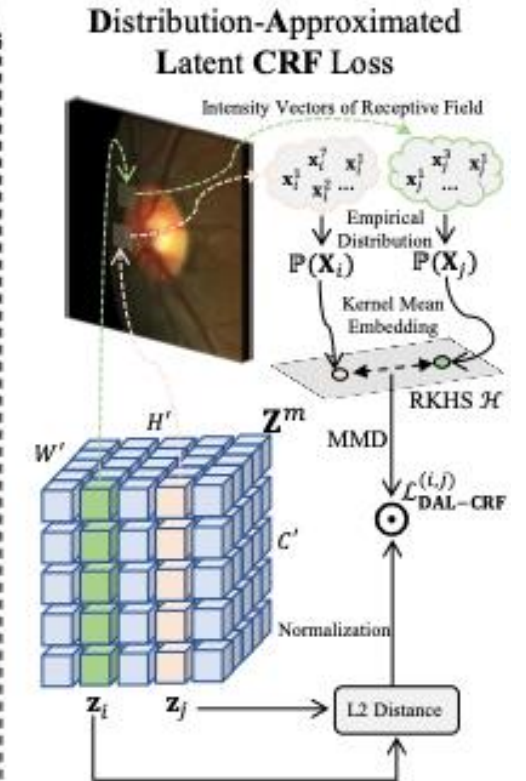
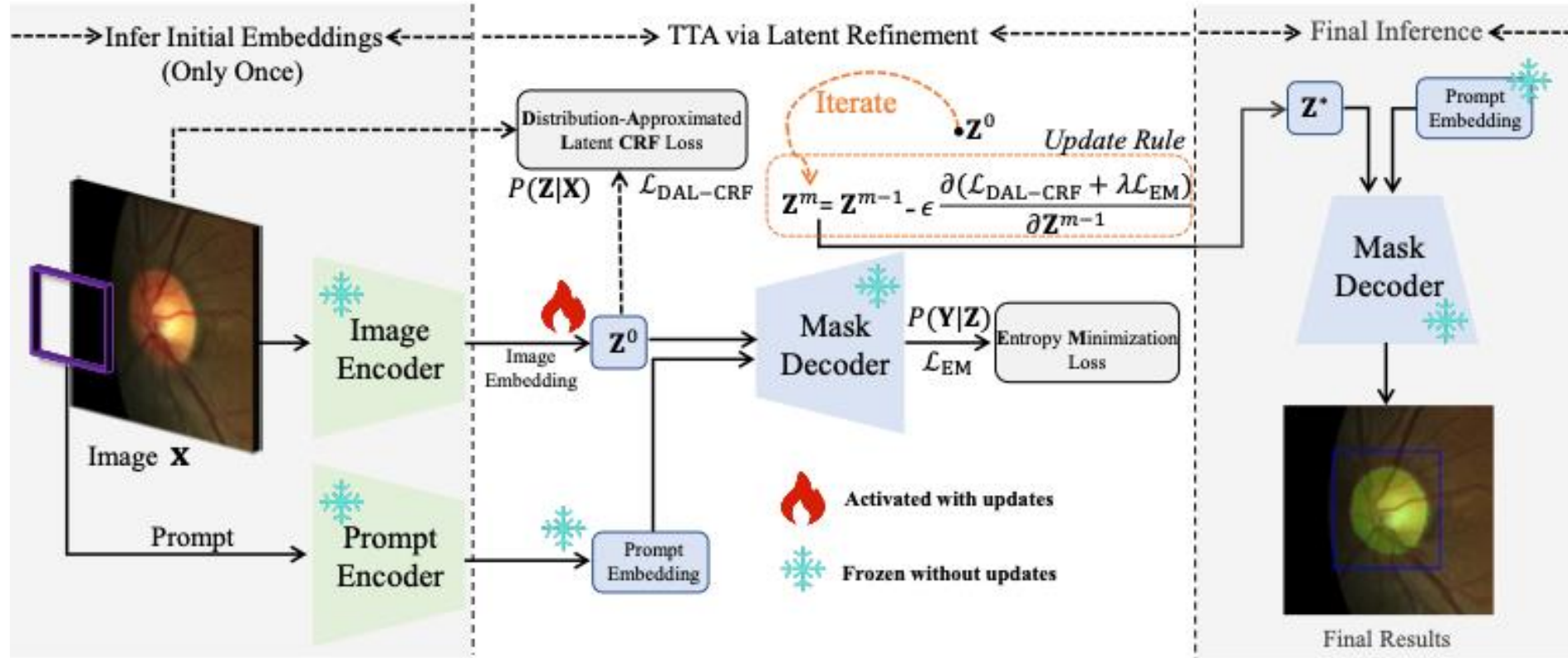
Latent Refinement for TTA

- Where latent refinement is commonly used: neural image compression
 - Reduce the bit consumption of the latent representation at compression time
- Why latent refinement is **more straightforward** in compression
 - An encoder–decoder architecture is trained to reconstruct the original raw image
 - The objective remains well-defined at test/compression time
- Why it remained “unexplored” for medical image segmentation TTA
 - There is **no direct-supervised signal**

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The framework of the proposed method



Why can we adapt by refining Z only?

- Definition
 - X : input medical image
 - Z : image embedding / latent representation produced by the encoder
 - Y : segmentation mask prediction
 - E_θ : encoder with parameters
 - : decoder with parameters φ
- Key idea:
 - $P(Y, X | Z) = P(Y | Z)P(X | Z)$

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Latent refinement instead of parametric update

- Eq. (1): formulate TTA as optimizing Z while freezing g, θ, φ

$$\mathbf{Y}^* = D_{\varphi}(\mathbf{Z}^*), \mathbf{Z}^* = \arg \max_{\mathbf{Z}} [\log P(\mathbf{Y} | \mathbf{Z}; \varphi) + \log P(\mathbf{Z} | \mathbf{X}; \theta)]$$

- Eq. (2): gradient-based latent update (generic form)

$$\mathbf{Z}^m = \mathbf{Z}^{m-1} - \epsilon \cdot \frac{\partial (-\log P(\mathbf{Y}^{m-1} | \mathbf{Z}^{m-1}) - \log P(\mathbf{Z}^{m-1} | \mathbf{X}))}{\partial \mathbf{Z}^{m-1}}$$

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Key loss 1: DAL-CR

- Eq. (3): each latent token corresponds to an input receptive field

$$\mathbf{z}_i \rightarrow \mathbf{X}_i = \{\mathbf{x}_i^l\}_{l=1}^L.$$

- Eq. (5): Maximum Mean Discrepancy (MMD) measures distribution discrepancy between two receptive fields.

$$\text{MMD}(P(\mathbf{X}_i), P(\mathbf{X}_j)) = \left\| \frac{1}{L} \sum_{l=1}^L \phi(\mathbf{x}_i^l) - \frac{1}{L} \sum_{l=1}^L \phi(\mathbf{x}_j^l) \right\|_{\mathcal{H}}^2.$$

Key loss 1: DAL-CR

- Eq. (4): DAL-CRF loss (distribution-aware latent CRF).

$$\mathcal{L}_{\text{DAL-CRF}} = \frac{1}{N'} \sum_{i,j \in \mathcal{G}} \left(\frac{1}{2} \text{MMD}(P(X_i), P(X_j)) \right) \cdot \exp \left(-\|\mathbf{z}_i - \mathbf{z}_j\|_{\mathbf{F}} \right).$$

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Key loss 2: Entropy Minimization

- Eq. (6): Entropy Minimization.

$$\mathcal{L}_{\text{EM}} = H(P(Y | Z)) = - \sum_{(h,v) \in C} P(y^{h,v} | Z) \log P(y^{h,v} | Z),$$
$$C = \{(h, v) \mid P(y^{h,v} | Z) \geq \alpha\}.$$

- Eq. (7): Final update rule.

$$\mathbf{Z}^m = \mathbf{Z}^{m-1} - \epsilon \cdot \frac{\partial(\mathcal{L}_{\text{DAL-CRF}} + \lambda \mathcal{L}_{\text{EM}})}{\partial \mathbf{Z}^{m-1}}.$$

The framework of the proposed method

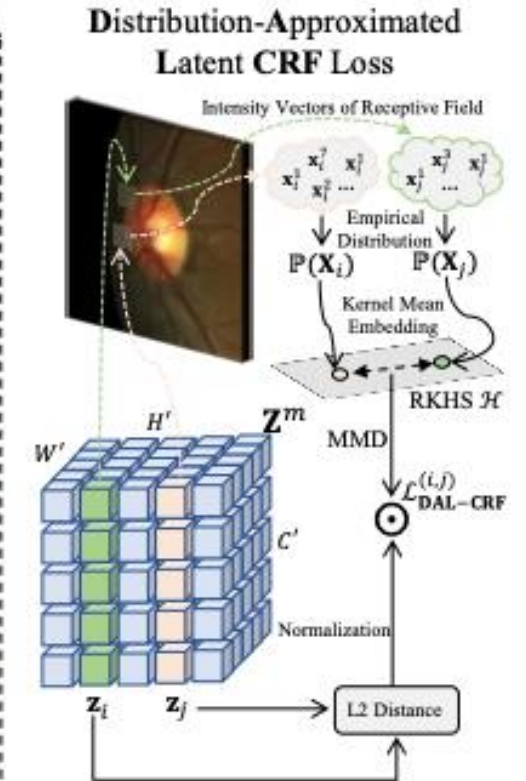
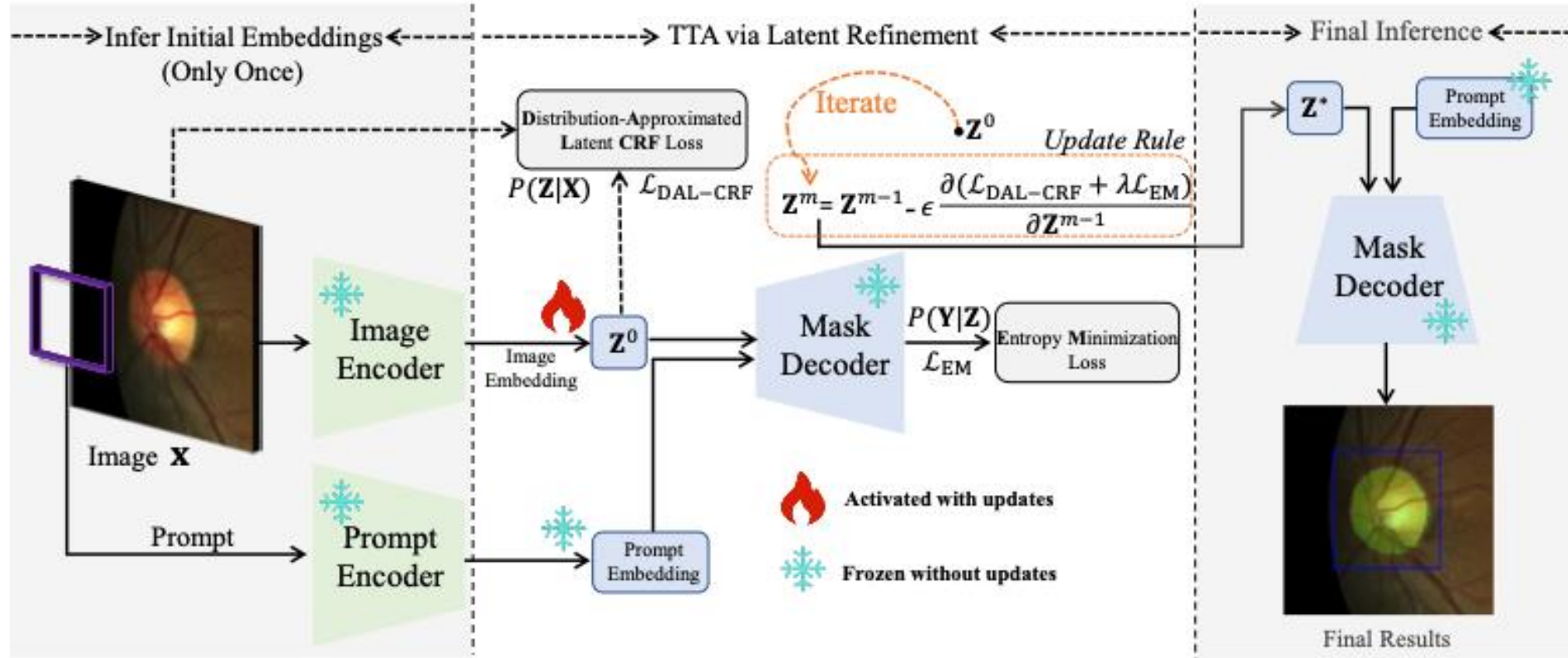


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Prompt perturbations

- The paper evaluates robustness to imperfect bounding-box prompts by constructing:
 - Perfectness prompt (PP): the reference box (e.g., derived from ground-truth mask).
 - Looseness prompt (LP): enlarge the box.
 - Shrinkage prompt (SP): shrink the box
- Why this matters.
 - In practice, prompt boxes can be loose or tight; performance can degrade under such perturbations.
 - The experiments quantify whether latent refinement can recover performance without updating model parameters.

Evaluation Matrix 1: Dice Coefficient

- P (Prediction), G (Ground-Truth)

$$\text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}.$$

- As confusion matrix

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}.$$

- Multi-class extension

$$\text{Dice}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{Dice}(P_c, G_c).$$

Evaluation Matrix 2: Average Surface Distance (ASD)

- P (Prediction), G (Ground-Truth)

$$\text{ASD}(\partial P \rightarrow \partial G) = \frac{1}{|\partial P|} \sum_{p \in \partial P} d(p, \partial G).$$

- Reverse Direction

$$\text{ASD}(\partial G \rightarrow \partial P) = \frac{1}{|\partial G|} \sum_{g \in \partial G} d(g, \partial P).$$

- Symmetric ASD

$$\text{ASD}(P, G) = \frac{1}{2} \left(\text{ASD}(\partial P \rightarrow \partial G) + \text{ASD}(\partial G \rightarrow \partial P) \right).$$

Table 2: Fundus OC Segmentation

Update Type	Method	Dice Coefficient $\uparrow$					Average Surface Distance $\downarrow$				
		D1	D2	D3	D4	Avg.	D1	D2	D3	D4	Avg.
Looseness prompts	Direct Inference	90.99	87.16	87.06	84.72	87.48	9.48	7.51	8.74	7.62	8.33
Normalization update	TENT (ICLR'21)	91.27	85.40	85.06	81.84	85.89	9.09	8.70	10.05	8.88	9.18
	InTEnt (CVPR'24)	91.36	<u>87.43</u>	<u>87.47</u>	<u>86.13</u>	87.60	8.80	<u>7.51</u>	<u>8.26</u>	7.66	8.06
	GraTa (AAAI'25)	90.89	87.00	86.95	84.64	87.37	9.48	7.54	8.80	<u>7.65</u>	8.37
	PASS (TMI'24)	80.86	79.63	82.03	84.02	81.63	20.40	12.22	11.69	7.97	13.07
Full updates	MEMO (NeurIPS'22)	87.26	85.48	86.42	84.47	85.91	12.45	8.27	9.15	7.76	9.40
	CRF-SOD (CVPR'23)	89.57	84.01	84.80	82.58	85.24	10.54	8.99	10.18	8.64	9.58
Latent updates	<i>Ours</i>	89.35	88.73	89.83	88.18	89.02	11.40	6.69	6.92	6.07	7.77
Shrinkage prompts	Direct Inference	81.05	67.60	72.10	68.57	72.33	18.51	17.61	17.76	14.92	17.20
Normalization update	TENT (ICLR'21)	80.22	65.17	69.25	62.10	69.18	18.85	19.26	19.39	17.58	18.77
	InTEnt (CVPR'24)	82.72	70.06	74.50	73.49	75.19	16.86	15.28	16.13	13.87	15.53
	GraTa (AAAI'25)	80.87	67.32	71.94	68.46	72.15	17.27	16.54	16.53	14.52	16.21
	PASS (TMI'24)	66.81	58.66	63.31	56.78	61.39	32.39	22.89	22.74	20.19	24.55
Full updates	MEMO (NeurIPS'22)	81.02	67.54	72.10	68.84	72.37	18.52	17.76	17.77	14.78	17.20
	CRF-SOD (CVPR'23)	81.84	69.26	74.73	72.08	74.48	19.27	18.83	18.63	14.73	17.86
Latent updates	<i>Ours</i>	83.07	73.72	78.54	75.77	77.77	16.74	14.38	13.72	11.26	14.02
Perfectness prompts	Direct Inference	87.58	80.25	81.32	78.46	81.90	12.76	11.21	12.27	10.24	11.62
Normalization update	TENT (ICLR'21)	87.60	77.98	78.18	72.99	79.18	<u>12.83</u>	12.21	13.92	12.69	12.91
	InTEnt (CVPR'24)	86.54	84.25	85.17	81.65	84.40	10.98	9.23	9.85	8.95	9.75
	GraTa (AAAI'25)	<u>87.39</u>	79.97	81.14	78.36	81.71	12.96	11.38	12.38	10.29	11.75
	PASS (TMI'24)	74.59	69.86	73.88	71.34	72.41	25.53	17.32	16.58	13.85	18.32
Full updates	MEMO (NeurIPS'22)	<u>87.39</u>	80.18	81.32	78.49	81.84	12.97	11.24	12.27	10.23	11.67
	CRF-SOD (CVPR'23)	86.07	77.22	79.10	76.24	79.65	13.96	12.59	13.63	11.2	12.84
Latent updates	<i>Ours</i>	86.85	85.11	86.72	84.49	85.79	13.73	8.63	8.79	7.48	9.65



Table 3: Fundus OD Segmentation

Update Type	Method	Dice Coefficient $\uparrow$					Average Surface Distance $\downarrow$				
		D1	D2	D3	D4	Avg.	D1	D2	D3	D4	Avg.
Looseness prompts	Direct Inference	83.05	78.86	81.82	75.54	79.81	33.20	32.87	30.29	31.90	32.06
Normalization update	TENT (ICLR'21)	81.61	76.96	80.00	74.85	78.35	37.58	36.54	33.53	34.35	35.50
	InTEnt (CVPR'24)	83.61	76.71	80.00	71.83	78.03	32.46	37.01	32.57	39.14	35.29
	GraTa (AAAI'25)	84.64	80.04	84.10	82.82	82.90	30.02	30.93	25.91	28.91	28.94
	PASS (TMI'24)	84.57	81.77	82.29	77.34	81.49	18.02	26.20	21.09	24.68	22.49
Full model updates	MEMO (NeurIPS'22)	86.26	80.01	83.94	78.18	82.10	26.73	30.93	26.27	28.54	28.12
	CRF-SOD (CVPR'23)	87.03	82.86	83.33	79.52	83.18	25.19	26.25	23.72	26.53	25.42
Latent updates	<i>Ours</i>	90.85	83.55	87.77	<u>82.24</u>	86.10	17.59	23.54	18.50	21.75	20.34
Shrinkage prompts	Direct Inference	92.70	86.17	90.90	88.67	89.61	12.60	17.66	12.77	12.10	13.78
Normalization update	TENT (ICLR'21)	90.63	85.63	89.66	86.88	88.20	15.90	18.44	14.27	12.48	15.27
	InTEnt (CVPR'24)	90.91	87.27	90.88	88.70	89.44	15.49	16.47	12.68	12.05	14.17
	GraTa (AAAI'25)	92.61	85.60	91.00	88.71	89.48	12.70	18.32	12.61	12.11	13.93
	PASS (TMI'24)	91.46	78.85	85.92	75.70	82.98	13.72	25.14	17.54	23.05	19.86
Full model updates	MEMO (NeurIPS'22)	92.12	83.82	90.87	88.77	88.89	13.52	20.18	12.75	12.04	14.62
	CRF-SOD (CVPR'23)	94.42	84.07	92.34	91.43	90.56	9.36	19.83	10.70	9.30	12.30
Latent updates	<i>Ours</i>	<u>94.24</u>	<u>86.77</u>	93.23	92.42	91.66	9.80	16.25	9.45	8.21	10.93
Perfectness prompts	Direct Inference	96.49	92.87	93.89	92.36	93.90	6.09	9.48	8.64	8.36	8.14
Normalization update	TENT (ICLR'21)	94.20	90.78	93.39	90.38	92.18	10.12	12.01	9.32	10.39	10.46
	InTEnt (CVPR'24)	94.12	91.57	94.42	93.47	93.39	10.12	11.01	8.01	7.03	9.04
	GraTa (AAAI'25)	94.46	92.66	93.91	92.31	93.33	6.14	9.73	8.61	8.42	8.22
	PASS (TMI'24)	94.90	85.35	84.36	86.14	87.68	8.58	17.52	17.62	13.98	14.42
Full model updates	MEMO (NeurIPS'22)	95.77	91.48	93.89	92.31	93.36	7.80	11.28	8.61	8.41	9.02
	CRF-SOD (CVPR'23)	96.66	91.15	93.76	93.41	93.75	5.78	11.42	8.73	7.26	8.29
Latent updates	<i>Ours</i>	96.68	92.67	95.15	94.57	94.76	5.69	9.51	6.85	5.98	7.00



Table 5/6: Ablation

(a)

K	D1	D2	D3	D4	Avg.
2	93.24	87.06	92.51	86.51	89.75
4	94.16	87.16	91.83	87.57	90.14
6	94.67	87.08	92.34	87.92	90.37
8	94.86	86.54	90.81	88.01	90.03

(b)

Loss	D1	D2	D3	D4	Avg.
DI	89.19	84.31	87.92	82.39	85.95
$\mathcal{L}_{\text{InTent}}$	79.34	84.89	88.83	86.08	84.78
$\mathcal{L}_{\text{TENT}}$	85.09	84.99	89.46	85.59	86.28
$\mathcal{L}_{\text{CRF-SOD}}$	88.93	86.17	88.42	87.33	87.71
$\mathcal{L}_{\text{DAL-CRF}}$	94.16	86.96	90.85	87.57	89.88

Method	D1	D2	D3	D4	Avg.
Direct Inference	89.19	84.31	87.92	82.39	85.95
+ $\mathcal{L}_{\text{EM}}$	85.82	85.31	88.10	83.21	86.01
+ $\mathcal{L}_{\text{DAL-CRF}}$	94.16	86.96	90.85	87.57	89.88
+ $\mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{DAL-CRF}}$	94.16	87.16	91.83	87.42	90.14
DenseCRF [11]-Postprocessing	82.70	79.39	80.90	73.76	79.18

Figure 2: Qualitative/Process Visualization

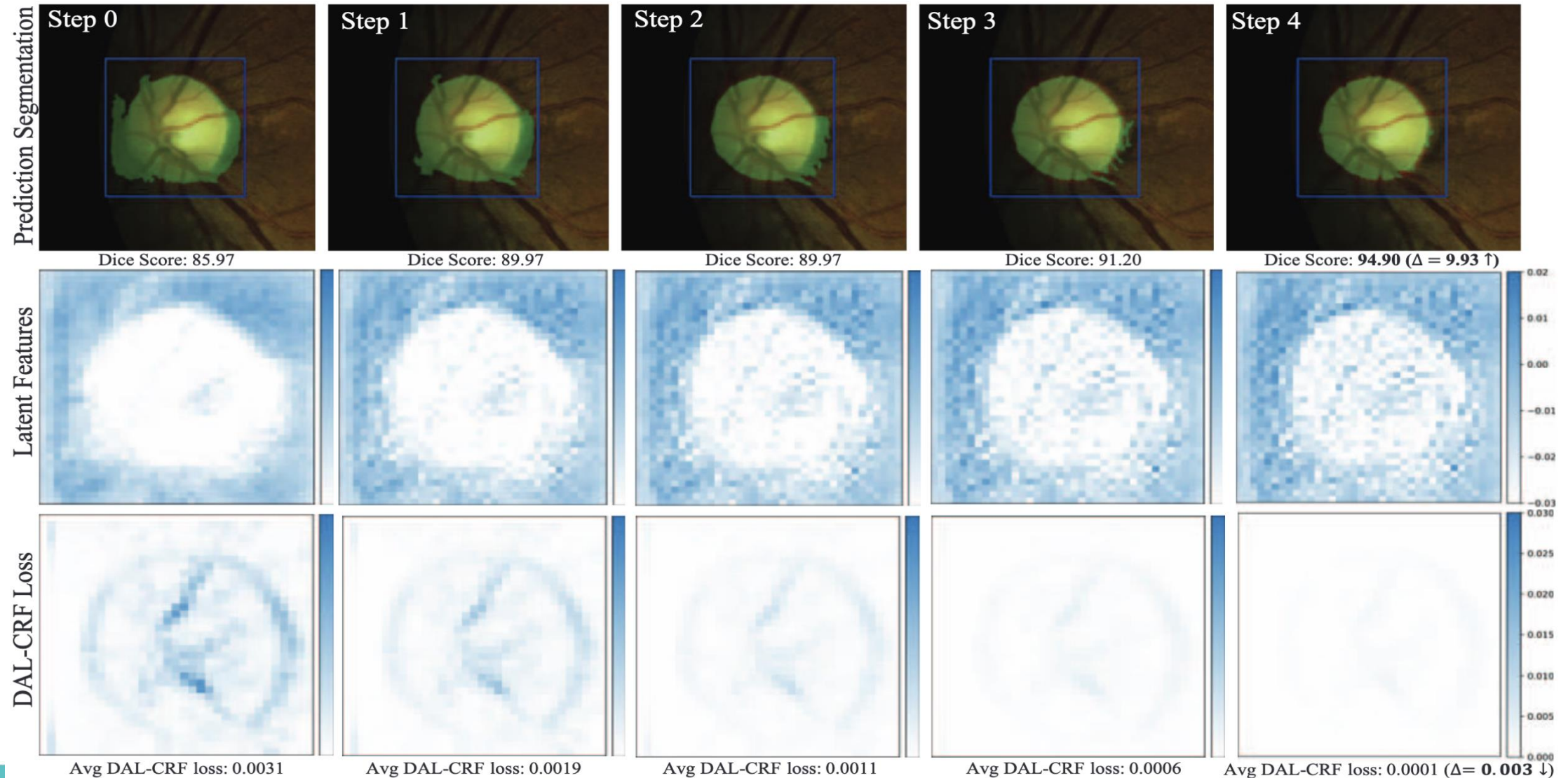


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Conclusion

- We propose a novel TTA framework for foundation medical segmentation without parametric updates. Our theoretical analysis shows that direct latent refinement is feasible within the MedSAM architecture. By maximizing factorized conditional probabilities by our DALCRF loss combined with entropy minimization, we achieve better performance for effectiveness and efficiency.

Thank You!